IoT-Enabled Smart Bus Route Monitoring & Revenue Management System



R26-ISE-004

Comments for the Proposal

Supervisor Comment	How we addressed them
1) Lack of creativity in presentation slides	Took steps to create more creative presentation slides within the scope
2) Too long individual components' deliveries	Practiced presentimg to ensure all parts are delivered within the given time
3) Not enough literiture review	Studied and read several other research papers to ensure well defined research gap

Problem, Impact & Overall Design

Why this system matters to transport authorities and bus owners

CORE PROBLEM:

The unavailibility of reliable public transport during off peak times

IMPACT:

causes severe physical and mental exhaustion and forces many to rely on expensive or unsafe travel alternatives

OTHER PROBLEMS:

1. Fixed permits fees miss income revenue opportunities
2. Lack of tracking causes route deviations
3. Manual schedule changes create uneven service gaps
4. Bus owners are unable to keep track of their drivers and work progress



BENEFICIARIES:

- **Passengers**: want more reliable public transportation
- **Transport authorit**y: want evidence based route monitoring and increased revenue
- **Bus owners**: want more transparent earnings
- **Bus drivers**: want clearer duty and schedule guidance



IT22228208|W S H R GUNASEKARA

Distance-based Revenue Management Component +Device

Distance-based Revenue Management Component +Device

How the solution addresses the scheduling and time-slot policy sub-problem

On-Board Gps device

Allows Location tracking+ Driver monitoring

Distance-based permit fee calculation model

More off-peak time service schedules

Mobile app contributions

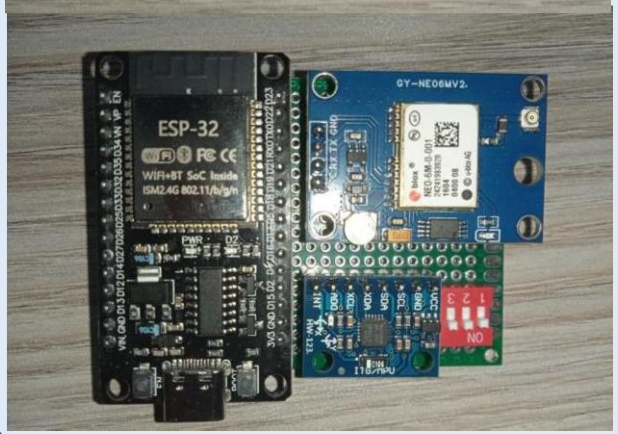
Permitted route violations



Distance-based Revenue Management Component +Device

DESIGN EXCELLENCE





Layer 01

- Perfboard
- 220µF 35V capacitor
- Glass Fuse
- Schottky Diode
- Resistors (SMD resistors)

Layer 02

- Perfboard
- ESP32 Development Board
- GY-NEO6MV2 GPS Module
- MPU6050 Sensor
- Micro USB / USB-C Port
- Crystal Oscillator

USER REQUIREMENTS ADDRESSED

- **Authorities**: Get increased permit fee revenue + Find route violations
- **Passengers**: Get a more reliable public transportation service
- **Bus owners**: Can increase income + keep track of their drivers
- **Bus drivers**: Increased salaries + operational guidance

PROTOTYPE



Policy-Driven Bus Scheduling and Time-Slot Management Engine



SACHINTHA H D D | IT22264152

BSc (Hons) in Information Technology Specializing in Information Systems Engineering

IT22264152 | Sachintha H D D | R26-ISE-004

Comparison chart of Research Gaps:

Feature	Proposed System	Research 1	Research 2	Research 3	Research 4
Configurable time-slot	✓	✗	✗	✗	✗
Policy-rule-based scheduling decisions	✓	✗	✗	✗	✗
schedule based on confirmed bus availability	✓	✗	✗	✗	✗
Real-time schedule/dispatch adjustment	✓	✓	✓	✗	✓

Research Papers:

1. **Zhang et al. (2021)** -An automatic real-time bus schedule redesign method based on bus arrival time prediction
2. **Dessouky et al. (2003)** – real-time control of buses for schedule coordination at a terminal
3. **Cevallos et al. (2011) -** Using AVL Data to Improve Transit On-Time Performance
4. **Rezazada et al. (2024)** - Bus bunching: a comprehensive review from demand

How Solution Assess Sub Problems:

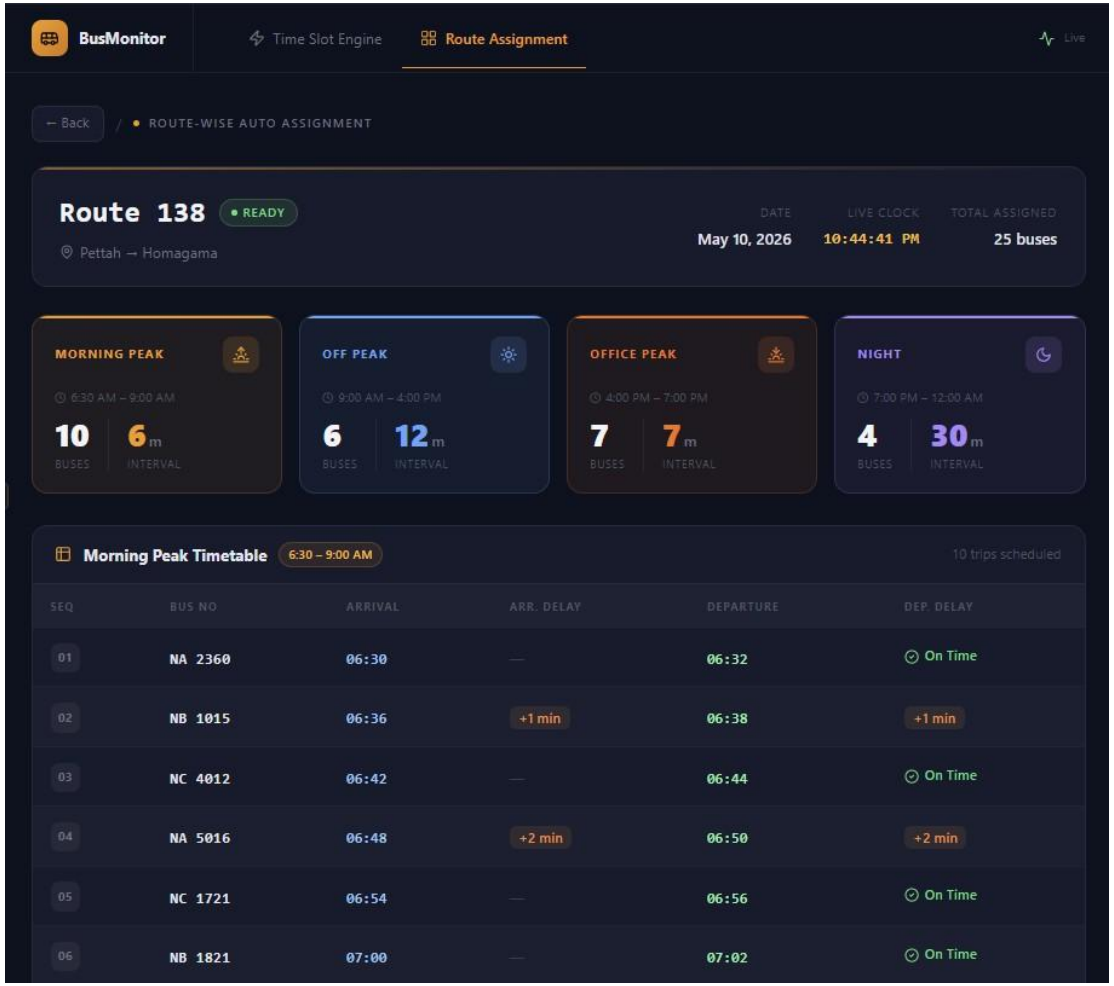
- **Fixed scheduling is inefficient:** Current systems cannot adapt when the actual number of available buses changes for the next day.

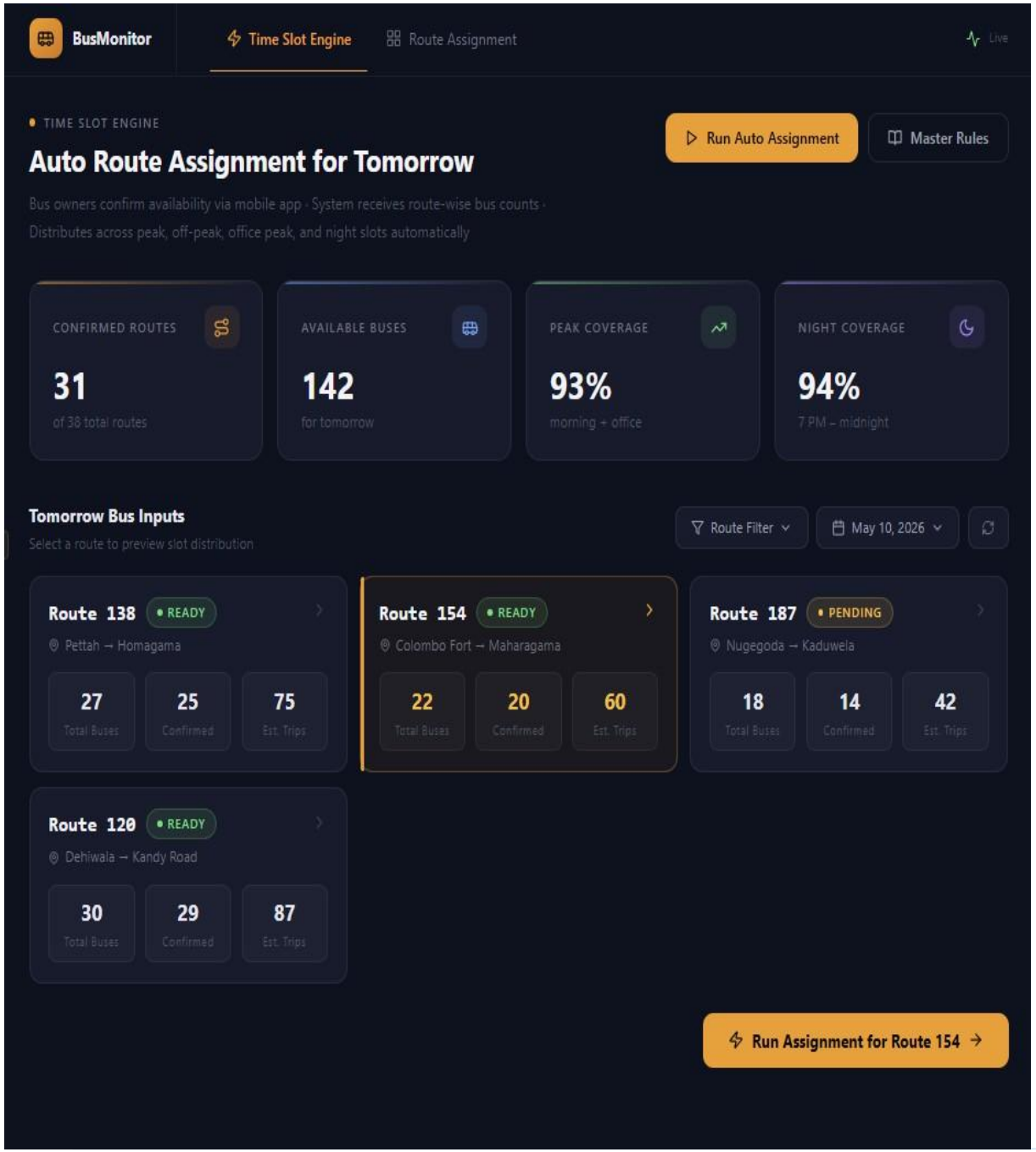
- **Weak route-wise planning:** Systems do not create the next day's route schedule according to the buses that are actually confirmed to run.

User Requirements Addressed by the Solution :

- Automatic time-slot distribution.
- Assignment visibility through dashboard tables.
- Policy rule configuration and updates.

Design Excellence:





Future Works:

- Front-end integration
- Mobile App and Web App Integration
- Add route-specific policy and threshold settings
- Develop delay penalty and grace-period handling rules.
- Add real-time visual panels for confirmed routes, available buses, peak coverage, and night coverage.

Event-Driven Bus Monitoring Engine

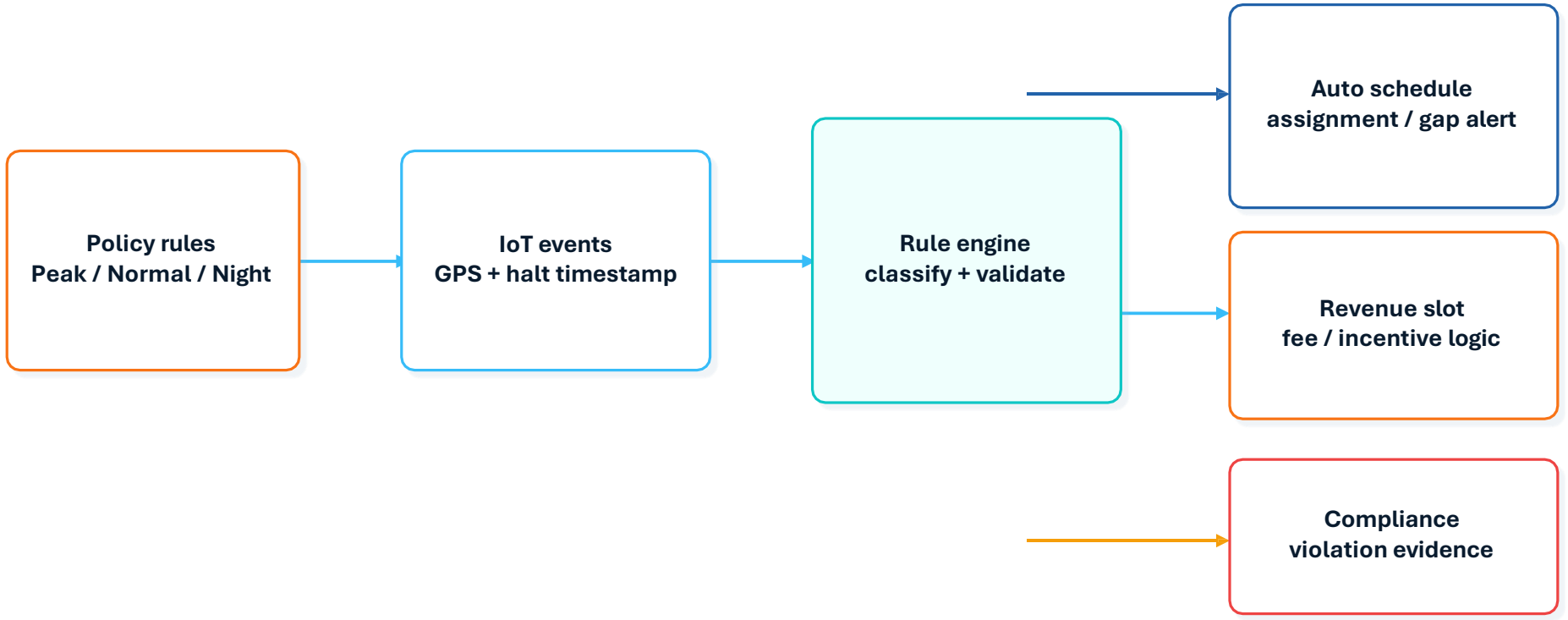


IT22121042 MIFLAL AHAMED M.T

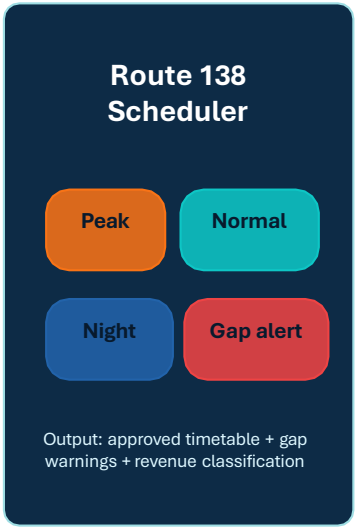
BSc (Hons) in Information Technology Specializing in Information Systems Engineering

Policy & Time-Slot Engine: Prototype Logic

How the solution addresses the scheduling and time-slot policy sub-problem



PROTOTYPE SCREENS



USER REQUIREMENTS ADDRESSED

Authority can configure route-wise peak, normal and night rules without changing code.
System must detect timetable gaps and suggest policy-compliant bus assignment.
Owner can view approved schedule, expected trips, payable fees and incentives.
Every decision keeps timestamped evidence for audit and dispute handling.

Design Excellence, Contribution & Prototype Feedback

What is technically new in the individual component



DESIGN CONTRIBUTION

Policy-first engine converts transport rules into executable scheduling logic.
Time-slot classification supports peak, normal and night operations with route-wise rates.
Disruption handling supports delayed buses, missed departures and bus count changes.
Versioned policy records give traceability for audits, penalties and owner disputes.

PROTOTYPE FEEDBACK / CHANGES MADE

Route selection was made faster and clearer for route officers.
Gap warnings were moved to the top of the schedule view.
Manual override was added for exceptional operational situations.
Revenue view was simplified into daily and monthly summaries.

IoT-Integrated Driver Monitoring and Operational Reporting Mobile Application Application



IT22925268 N.D.ATHTHANAYAKA

BSc (Hons) in Information Technology Specializing in Information Systems Engineering

Sources: Ministry of Transport statistics page; National Transport Commission Annual Report 2023; Western Province route information PDF.

Comparison chart of Research Gaps:

Feature	Proposed System	Research 1	Research 2	Research 3	
Real-time driver behavior monitoring	✓	✓	✗	✗	
Gyroscope-based behavior analysis	✓	✗	✗	✗	
Aggressive braking, turning, Abnormal slow movement detection	✓	✗	✗	✗	
Instant alerts to driver	✓	✓	✗	✓	
Mobile app support, Owner-side trip summary view	✓	✗	✗	✗	

Research Papers:

1. **Research (1)** = *A Comprehensive Review of Deep Learning and IoT in Driver Drowsiness Detection for Safer Roads*
2. **Research (2)** = *Digital-Twin-Driven Performance Assessment for IoT-Integrated Autonomous Driving Systems*
3. **Research (3)** = *IoT Based Intelligent Bus Monitoring System*

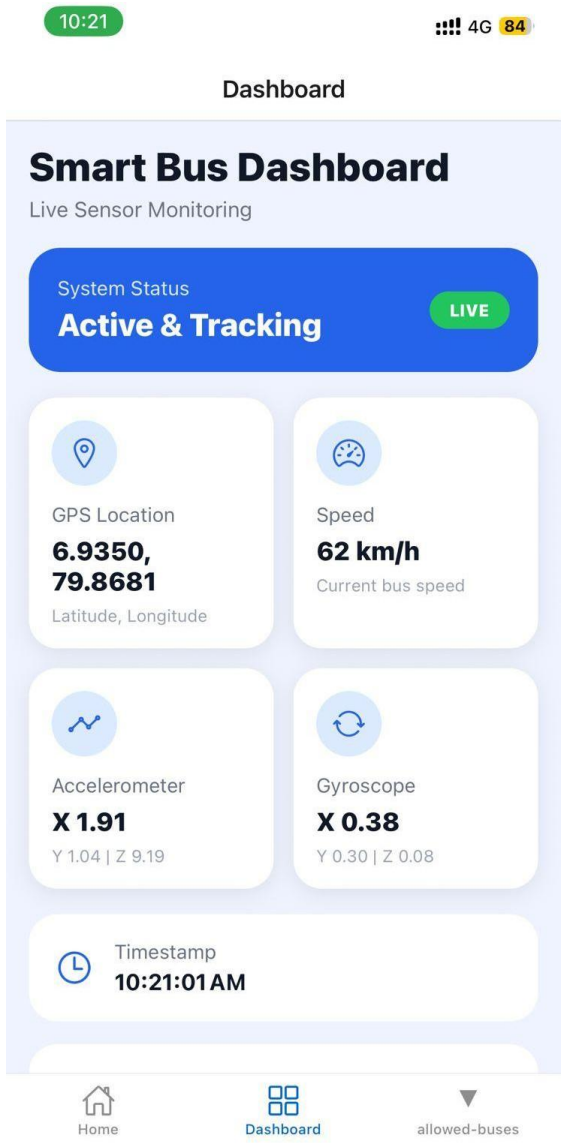
How Solution Assess Sub Problems:

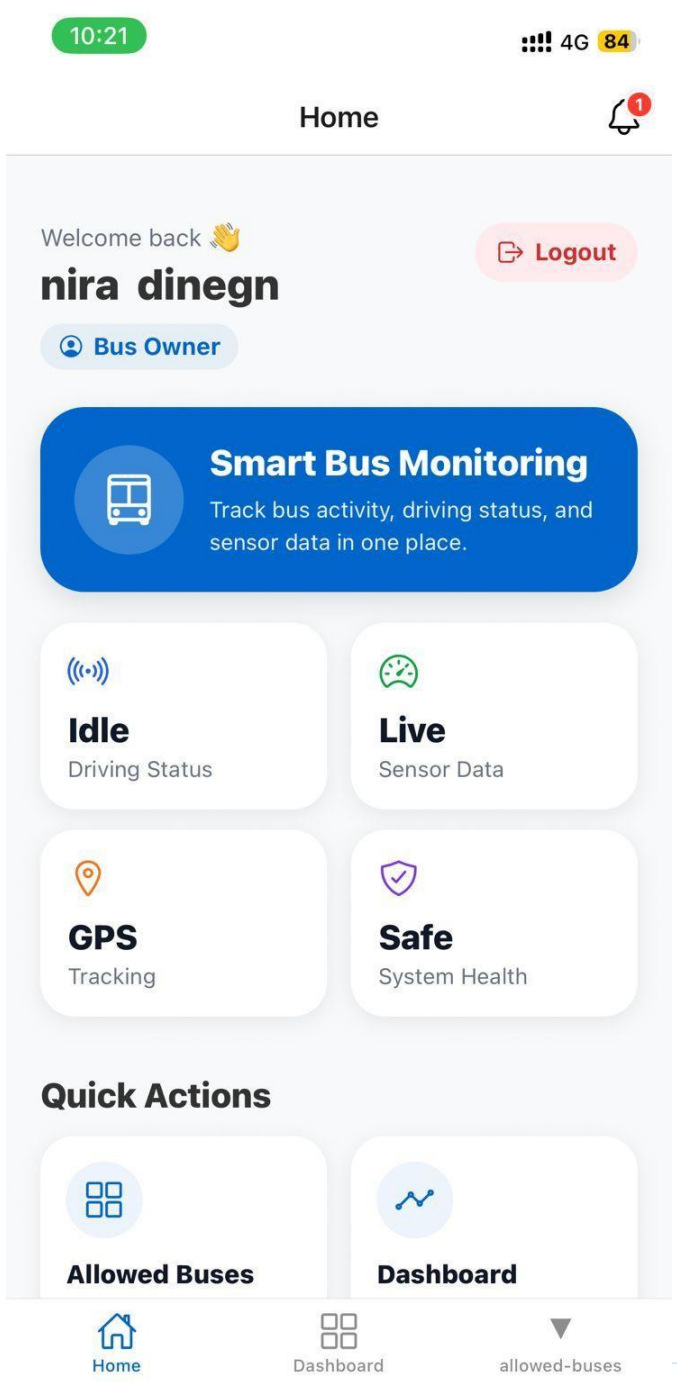
- **Unsafe driving is difficult to detect in real time:** Existing bus systems mainly focus on location or general tracking, but they do not continuously identify unsafe driving behaviors such as overspeeding, aggressive braking, aggressive turning, and abnormal slow movement.
- **No immediate feedback to the driver:** Current methods do not provide instant warnings when unsafe driving events happen, so drivers may continue risky behavior without awareness..
- **Limited owner visibility into driver performance:** Bus owners cannot easily monitor whether the driver is operating the bus safely or review real-time driving behavior during daily operations.
- **Weak operational reporting through mobile platforms:** Existing systems provide limited support for mobile-based trip summaries, driver behavior reports, bus availability updates, breakdown reporting, schedules, earnings, and route-related payment visibility.

User Requirements Addressed by the Solution :

- Real-time monitoring of driver behavior using onboard sensor data..
- Instant mobile alerts for unsafe driving events..
- Trip summary and driver behavior report visibility for bus owners..
- Bus availability and trip status reporting through the mobile app.
- Breakdown notification support for daily operations..

Design Excellence:





Future Works:

- Improve real-time detection accuracy for unsafe driving events.
- Enhance mobile app integration with web dashboard reporting.
- Extend trip summaries and driver performance analytics
- Develop accident-risk prediction and predictive alerts.

Design Excellence, Contribution & Prototype Feedback

What is technically new in the individual component



DESIGN CONTRIBUTION

Policy-first engine converts transport rules into executable scheduling logic.
Time-slot classification supports peak, normal and night operations with route-wise rates.
Disruption handling supports delayed buses, missed departures and bus count changes.
Versioned policy records give traceability for audits, penalties and owner disputes.

PROTOTYPE FEEDBACK / CHANGES MADE

Route selection was made faster and clearer for route officers.
Gap warnings were moved to the top of the schedule view.
Manual override was added for exceptional operational situations.
Revenue view was simplified into daily and monthly summaries.

Customer Persona



User Persona: Transport Authority Senior Operations Manager

W.M.I.K. Wijerathnayake

Age: 53

Gender: Male

Location: No. 140/5, Udupila, Delgoda

Occupation: Deputy General Manager

Mr. Wijerathnayake is a senior operations officer responsible for supervising public bus operations within the Western Province. His work involves managing route operations, monitoring service availability, handling operational issues, coordinating with bus owners and transport officers, and supporting policy-based decisions.

Background

Goals

- Improve public bus availability and reliability across approved routes.
- Monitor whether buses operate according to transport authority rules.
- Reduce unauthorized route deviations.
- Improve off-peak and night-time bus service coverage.

Needs

- A clear authority dashboard showing route-wise bus activity.
- Real-time or event-based bus monitoring data.
- Route deviation and non-compliance alerts.
- Bus availability and breakdown reports.
- Automated scheduling support during delays or breakdowns.

Frustrations

- Manual bus schedules are difficult to monitor in real time.
- Route deviations may be discovered only after complaints or inspections
- Bus owners may provide incomplete or delayed operational information.
- Fixed permit systems do not always reflect actual bus usage.

Commercialization & Sustainability

MARKET SIZE

Western Province had 6,082 bus fleet and 2,621 operated buses in 2021; NTC reported about 3,200 inter-provincial buses on 450 routes in 2023.

UNIQUE VALUE

This system provides not only live vehicle tracking but converts it into a revenue-generating business model that serves both the government and its citizens.

HOW COST WILL BE RECOVERED

The project cost can be recovered through IoT device installation charges, monthly maintenance subscriptions, and service agreements with transport authorities and bus operators. Since the system helps authorities increase permit revenue and reduce manual monitoring costs, the financial benefits gained from improved transport management can support the long-term sustainability of the project.

SUSTAINABILITY

The ability to improve public transport efficiency using low-cost IoT technology and data-driven management unlike existing solutions makes this system technicaly sustainble while the distance based permit model ensures fair and continuous government revenue, making the system also financially sustainable.